Gideon A. Afriyie

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Education

Gettysburg College *Gettysburg, PA - USA*

B.A Mathematics, Minor in Economics *2023 - 2027*

- Honors: AMS–Math Alliance F-GAP Scholar; Taylor Gaw '13 Endowed Fellowship; Seibert Project Award

Experience

Advancing Science Lab *On-campus*

Research Assistant – 3D Topographic Modeling *Aug 2026 – Present*

- Developing 3D topographic models using elevation data and tools including TouchTerrain, QGIS, and UltiMaker Cura.
- Testing model scale, vertical exaggeration, base dimensions, and print settings through multiple prototype iterations.
- Exploring methods for incorporating roads, waterways, and the Appalachian Trail using GIS data.

Under Canvas Inc. *Remote*

Data Analytics Fellow *May – Jul 2025*

- Analyzed a $1.6M organizational compensation dataset using Python and Excel to identify anomalies across employee groups.
- Developed quantitative recommendations for incentive restructuring; findings presented to leadership.

Brighter Generation

Volunteer Teaching Assistant *May – Sept. 2024*

- Mentored 25+ high school students, supporting learning activities, student projects, and development.

Publication

Published

Afriyie, Gideon A. (2026). Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér's Conjecture and Light-Tailed Behavior. *Statistics & Probability Letters*. DOI: 10.1016/j.spl.2026.110898

Projects

Quantitative Risk & Portfolio Optimization | GitHub

- Analyzed 6+ years of daily prices across 8 assets, estimating returns, volatility, and correlations, including -0.754 VOO–VIXY correlation.
- Constructed efficient frontiers and optimal portfolios using mean-variance optimization; applied KKT conditions to enforce no-short-selling constraints.

Skills

Python, Excel, R, SQL, LaTeX

Research & Programs

2026 **Probability for AI**, Stanford University *Oct–Nov 2026*

2026 **TPU Research Cloud**, Google *May 2026*

2026 **AI Summer School**, MIT/Harvard IAIFI *August 2026*